HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented Python scripts to automate ETL processes for data ingestion into enterprise data warehouses, reducing manual processing time by approximately **40%**.
- Developed and maintained Java Spring Boot REST APIs for inter-service communication, handling over **3,500 requests per minute** across multiple microservices.
- Managed production support for enterprise applications on AWS (EC2, S3, Lambda) using Splunk and Dynatrace for monitoring and incident response, ensuring adherence to critical SLAs.
- Contributed to highway design calculations and technical drawing production using industry-standard software, focusing on junction improvements and active travel routes.
- Analyzed traffic and transport data to identify patterns and inform design decisions for road network upgrades, supporting sustainable transport solutions.
- Collaborated with multidisciplinary project teams to develop practical solutions for complex engineering challenges within transport infrastructure projects.
- Containerized Java Spring Boot applications using **Docker** and deployed them to an **AWS EKS** cluster, improving deployment consistency and enabling easier scaling.

SKILLS

Languages – Python, Java, SQL, R

Frameworks/Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Kubernetes

Databases – PostgreSQL, MySQL, Oracle

Tools/Platforms – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau

Engineering Concepts – REST APIs, ETL, CI/CD, Agile Methodologies, Data Analysis, Transport Networks, Infrastructure Design

PROJECTS

Automated Data Pipeline for Infrastructure Analytics

- Developed a Python-based ETL pipeline to ingest and transform diverse data sources, including sensor readings and traffic flow logs, for downstream analytical processing.
- Implemented data validation and cleansing routines within the pipeline, ensuring data integrity for over **1 TB** of processed information weekly.
- Integrated the pipeline with a PostgreSQL database, structuring data for efficient querying by BI tools and analytical models focused on transport network performance.

Java Microservice for Transport Network Monitoring

- Engineered a Java Spring Boot microservice to process real-time data streams from road sensors and traffic management systems, contributing to network operational awareness.
- Designed and implemented RESTful endpoints for data ingestion and status reporting, supporting integration with existing enterprise monitoring platforms.
- Wrote comprehensive JUnit tests and integrated Mockito for unit and integration testing, achieving over **90% code coverage** for critical service components.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)